Visualizing Relationships

1. **Scatterplot with Regression Line:**

```
sns.regplot(x='x', y='y', data=df, scatter_kws={'color': 'red'}, line_kws={'color': 'blue'})
plt.title('Scatterplot with Regression Line')
plt.show()
```

2. **Pairplot:**

Pairplots provide pairwise relationships for numerical variables in a dataset:
sns.pairplot(df)

plt.show()

Heatmaps

Heatmaps are effective for visualizing correlations or matrix data:
corr_matrix = df.corr()

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm') plt.title('Correlation Heatmap')
plt.show()

Combining Matplotlib and Seaborn

Using Matplotlib and Seaborn together can lead to highly customized visualizations.

For example:

```
sns.boxplot(x='category', y='value', data=df)
plt.title('Customized Box Plot')
plt.xlabel('Categories')
plt.ylabel('Values')
plt.grid(True)
plt.show()
```

Best Practices for Data Visualization

1. **Choose the Right Chart Type:** Ensure that the chart type matches the nature of your data and the insights you wish to convey.
2. **Keep It Simple:** Avoid overloading charts with too many elements; focus on clarity.
3. **Label Axes and Titles Clearly:** Provide meaningful labels and titles to help viewers interpret the visualization.